Mat 214 Nümerik Yöntemler proje ödevi4

1)

$$\left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 8 \\ 2 & -1 & -1 & 1 & 5 \\ 2 & 2 & 1 & 0 & 4 \\ 2 & -1 & 2 & 2 & 10 \end{array}\right] \xrightarrow{(-1)\times 1. + 4. = 4.} \left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 8 \\ 2 & -1 & -1 & 1 & 5 \\ 2 & 2 & 1 & 0 & 4 \\ 0 & 0 & 0 & 2 & 2 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 8 \\ 2 & -1 & -1 & 1 & 5 \\ 2 & 2 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow{-4+2.=2} \left[\begin{array}{cccc|c} 2 & -1 & 2 & 0 & 8 \\ 2 & -1 & -1 & 0 & 4 \\ 2 & 2 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right]$$

$(-1)\times 3. + 1. = 1.$
$(-1)\times 3. + 2. = 2.$

$$\Rightarrow \left[\begin{array}{cccc|c} 0 & -3 & 1 & 0 & 4 \\ 0 & -3 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow{(-1)\times 2. + 1. = 1.} \left[\begin{array}{cccc|c} 0 & 0 & 3 & 0 & 4 \\ 0 & -3 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right]$$

$3x_3 = 4$

$\boxed{x_3 = \frac{4}{3}}$ $\qquad \boxed{x_4 = 1}$

$x_3 = 1,333$

$-3x_2 - 2x_3 = 0$

$3x_2 = -2x_3$

$3x_2 = -2 \cdot \frac{4}{3}$

$\boxed{x_2 = -\frac{8}{9} = -0,8889}$

$2x_1 + 2x_2 + x_3 = 4$

$2x_1 - \frac{16}{9} + \frac{4}{3} = 4$

$2x_1 - \frac{4}{9} = 4$

$\boxed{x_1 = 2 + \frac{2}{9} = 2,2222}$

2)

$$\begin{bmatrix} 2 & 3 & -1 \\ 1 & -2 & 2 \\ -1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 5 \end{bmatrix}$$

$$x_1 = \frac{|A_1|}{|A|} = \frac{-62}{-31} = 2$$

$$x_2 = \frac{|A_2|}{|A|} = \frac{-31}{-31} = 1$$

$$x_3 = \frac{|A_3|}{|A|} = -\frac{93}{-31} = 3$$

3) LU faktorizasyonu yardımıyla sistemi çözünüz.

$$\underbrace{\begin{bmatrix} 2 & -1 & 1 \\ 3 & 3 & 9 \\ 3 & 3 & 5 \end{bmatrix}}_{A} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 3 & 3 & 9 \\ 3 & 3 & 5 \end{bmatrix} \xrightarrow{2.-\left[\left(\frac{3}{2}\right)\cdot 1.\right]=2.} \begin{bmatrix} 2 & -1 & 1 \\ 0 & \frac{9}{2} & \frac{15}{2} \\ 3 & 3 & 5 \end{bmatrix} \xrightarrow{3.-\left(\frac{3}{2}\right)\cdot 1.=3.} \begin{bmatrix} 2 & -1 & 1 \\ 0 & \frac{9}{2} & \frac{15}{2} \\ 0 & \frac{5}{2} & \frac{7}{2} \end{bmatrix}$$

$$\xrightarrow{3.-2.=3.} \begin{bmatrix} 2 & -1 & 1 \\ 0 & \frac{9}{2} & \frac{15}{2} \\ 0 & 0 & -\frac{8}{2} \end{bmatrix} = U = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 4.5 & 7.5 \\ 0 & 0 & -4 \end{bmatrix}$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ \frac{3}{2} & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1,5 & 1 & 0 \\ 1,5 & 1 & 1 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ \frac{3}{2} & 1 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & -1 & 1 \\ 0 & 9/2 & \frac{15}{2} \\ 0 & 0 & -4 \end{bmatrix}}_{U} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$$

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ \frac{3}{2} & 1 & 0 \\ 3/2 & 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 4 \end{bmatrix}$$

$a = -1$

$\frac{3}{2}a + b = 0$

$b = -\frac{3}{2}a = \frac{3}{2}$

$\frac{3}{2}a + b + c = 4$

$c = 4$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 9/2 & \frac{15}{2} \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ \frac{3}{2} \\ 4 \end{bmatrix}$$

$-4x_3 = 4$

$\boxed{x_3 = -1}$

$\frac{9}{2}x_2 + \frac{15}{2}x_3 = \frac{3}{2}$

$\frac{9}{2}x_2 = \frac{18}{2} = 9$

$\boxed{x_2 = 2}$

$2x_1 = x_2 - x_3 - 1$

$= 2 + 1 - 1 = 2$

$\boxed{x_1 = 1}$

$c = -\frac{19}{5} \cdot \frac{\sqrt{5}}{\sqrt{19}} = -\frac{\sqrt{19}\cdot\sqrt{19}\cdot\sqrt{5}}{\sqrt{5}\cdot\sqrt{5}\cdot\sqrt{19}} = -\frac{\sqrt{19}}{\sqrt{5}}$

4)

a) $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 6 & 2 \\ 2 & 2 & 5 \end{bmatrix}$

i) $x^T = [x_1, x_2, x_3]$

$$x^T A x = [x_1, x_2, x_3] \begin{bmatrix} 4 & 2 & 2 \\ 2 & 6 & 2 \\ 2 & 2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= [x_1, x_2, x_3] \begin{bmatrix} 4x_1 + 2x_2 + 2x_3 \\ 2x_1 + 6x_2 + 2x_3 \\ 2x_1 + 2x_2 + 5x_3 \end{bmatrix}$$

$$= 4x_1^2 + 2x_1x_2 + 2x_1x_3 + 2x_1x_2 + 6x_2^2 + 2x_3x_2 + 2x_1x_3 + 2x_2x_3 + 5x_3^2$$

$$= 4x_1^2 + 4x_1x_2 + 6x_2^2 + 4x_1x_3 + 5x_3^2 + 4x_2x_3$$

$$= x_1^2 + 4x_1x_2 + 4x_2^2 + 3x_1^2 + 2x_2^2 + 4x_1x_3 + 5x_3^2 + 4x_2x_3$$

$$= (x_1 + 2x_2)^2 + (\sqrt{2}x_1 + \sqrt{2}x_3)^2 + x_1^2 + 2x_2^2 + 2x_3^2 + 4x_2x_3$$

$$= (x_1 + 2x_2)^2 + (\sqrt{2}x_1 + \sqrt{2}x_3)^2 + (\sqrt{2}x_2 + \sqrt{2}x_3)^2 + (x_1)^2$$

$$> 0$$

ii) $A_1 = 4 \quad \det(A_1) = 4$

$A_2 = \begin{vmatrix} 4 & 2 \\ 2 & 6 \end{vmatrix} \quad \det(A_2) = 20$

$A_3 = A \quad \det(A_3) = 76$

hepsi pozitiftir

4) b) $A = R^T \cdot R$

$$A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 6 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

$$A = \begin{bmatrix} r_{11} & 0 & 0 \\ r_{12} & r_{22} & 0 \\ r_{13} & r_{23} & r_{33} \end{bmatrix} \cdot \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{bmatrix} = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 6 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

$r_{11}^2 = 4$ $\quad r_{11} \cdot r_{12} = 2$ $\quad r_{11} \cdot r_{13} = 2$

$r_{11} = 2$ $\quad r_{12} = 1$ $\quad r_{13} = 1$

$r_{12} \cdot r_{22} = 2$ $\quad (r_{12})^2 + (r_{22})^2 = 6$

$r_{22} = \sqrt{5} = 2,2361$

$\frac{r_{12}}{1} \quad \frac{r_{13}}{1} + \frac{r_{22} \cdot r_{23}}{\sqrt{5}} = 2$

$r_{23} = \frac{1}{\sqrt{5}} = 0,4472$

$\frac{r_{13}^2}{1} + \frac{r_{23}^2}{\frac{1}{5}} + r_{33}^2 = 5$

$r_{33}^2 = 4 - \frac{1}{5} = \frac{19}{5}$

$r_{33} = \sqrt{\frac{19}{5}} = 1,9494$

$$R = \begin{bmatrix} 2 & 1 & 1 \\ 0 & \sqrt{5} & \frac{1}{\sqrt{5}} \\ 0 & 0 & \sqrt{\frac{19}{5}} \end{bmatrix}$$

c) $Ax = b \Rightarrow R^T \cdot R x = b$

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & \sqrt{5} & 0 \\ 1 & \frac{1}{\sqrt{5}} & \sqrt{\frac{19}{5}} \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & \sqrt{5} & \frac{1}{\sqrt{5}} \\ 0 & 0 & \sqrt{\frac{19}{5}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$2a = 4$

$a = 2$

$a + \sqrt{5} b = 6$

$\frac{\sqrt{5} b}{\sqrt{5}} = \frac{4}{\sqrt{5}}$

$b = \frac{4}{\sqrt{5}}$

$a + \frac{b}{\sqrt{5}} + \sqrt{\frac{19}{5}} c = -1$

$2 + \frac{4}{5} + \sqrt{\frac{19}{5}} c = -1$

$\sqrt{\frac{19}{5}} c = -3 - \frac{4}{5} = -\frac{19}{5}$

$c = -\frac{19}{5} \cdot \frac{\sqrt{5}}{\sqrt{19}} = -\frac{\sqrt{19} \cdot \sqrt{19} \cdot \sqrt{5}}{5 \cdot \sqrt{19}} = -\frac{\sqrt{19}}{\sqrt{5}}$

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & \sqrt{5} & \frac{1}{\sqrt{5}} \\ 0 & 0 & \sqrt{\frac{19}{5}} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ \frac{4}{\sqrt{5}} \\ -\sqrt{\frac{19}{5}} \end{bmatrix}$$

$\sqrt{\frac{19}{5}} x_3 = -\sqrt{\frac{19}{5}}$

$x_3 = -1$

$\sqrt{5} x_2 + \frac{x_3}{\sqrt{5}} = \frac{4}{\sqrt{5}}$

$\sqrt{5} x_2 = \frac{5}{\sqrt{5}}$

$x_2 = 1$

$2x_1 + x_2 + x_3 = 2$

$2x_1 + 0 = 2$

$\boxed{x_1 = 1}$

5) a) $A = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$ $\quad Ax = \lambda x$

$$(A - \lambda I) = \begin{bmatrix} 1-\lambda & 2 \\ -1 & 2-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (1-\lambda)(2-\lambda) - (2)(-1)$$

$$= 2 - \lambda - 2\lambda + \lambda^2 + 2$$

$$= \lambda^2 - 3\lambda + 4 = 0$$

$$\Delta = b^2 - 4ac = 9 - 4 \cdot 1 \cdot 4 = -7$$

$$x_1 = \frac{-b + \sqrt{\Delta}}{2a} = \frac{+3 + \sqrt{-7}}{2} = \frac{3 + \sqrt{7}\,i}{2}$$

$$x_2 = \frac{3 - \sqrt{-7}}{2} = \frac{3 - \sqrt{7}\,i}{2}$$

$$D = \begin{bmatrix} \frac{3+\sqrt{7}\,i}{2} & 0 \\ 0 & \frac{3-\sqrt{7}\,i}{2} \end{bmatrix}$$

spektrum yarıçapı

$\max\{|\lambda_1|, |\lambda_2|\} = \frac{3+\sqrt{7}}{2}$

$\lambda_1 = \frac{3+\sqrt{7}\,i}{2}$ için

$$\begin{bmatrix} \frac{-1-\sqrt{7}\,i}{2} & 2 \\ -1 & \frac{1+\sqrt{7}\,i}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-\frac{1+\sqrt{7}\,i}{2} x_1 + 2x_2 = 0$$

$$x_2 = \frac{1+\sqrt{7}\,i}{4} x_1 \qquad \begin{bmatrix} x_1 \\ \frac{1+\sqrt{7}\,i}{4} x_1 \end{bmatrix}$$

$\lambda_2 = \frac{3-\sqrt{7}\,i}{2}$

$$\begin{bmatrix} 1 - \frac{3-\sqrt{7}\,i}{2} & 2 \\ -1 & 2 - \frac{3-\sqrt{7}\,i}{2} \end{bmatrix} = \begin{bmatrix} \frac{-1+\sqrt{7}\,i}{2} & 2 \\ -1 & \frac{1+\sqrt{7}\,i}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$-\frac{1+\sqrt{7}\,i}{2} x_1 + 2x_2 = 0$$

$$x_2 = \frac{-1+\sqrt{7}\,i}{4} \qquad \begin{bmatrix} x_1 \\ \frac{-1+\sqrt{7}\,i}{4} x_1 \end{bmatrix}$$

5) b)

$$A = \begin{bmatrix} -1 & 2 & 0 \\ 0 & 3 & 4 \\ 0 & 0 & 7 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} -1-\lambda & 2 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 0 & 7-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(-1-\lambda)\begin{vmatrix} 3-\lambda & 4 \\ 0 & 7-\lambda \end{vmatrix} - 2\begin{vmatrix} 0 & 4 \\ 0 & 7-\lambda \end{vmatrix}$$

$$= (-1-\lambda)(3-\lambda)(7-\lambda)$$

$-1 = \lambda_1$ $\lambda_2 = 3$ $\lambda_3 = 7$

$\lambda_1 = -1$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 4 & 4 \\ 0 & 0 & 8 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$2x_2 = 0$ $\quad x_3 = 0$ $\quad 4x_1 + 4x_2 = 0$

$x_2 = 0$ $\quad x_1 = -x_2$

$$\begin{bmatrix} x_1 \\ 0 \\ 0 \end{bmatrix}$$

$\lambda_2 = 3$

$$\begin{bmatrix} -4 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 4 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$-4x_1 + 2x_2 = 0$ $\quad x_3 = 0$ $\quad \begin{bmatrix} x_1 \\ 2x_2 \\ 0 \end{bmatrix}$

$2x_2 = 4x_1$ $\quad x_2 = 2x_1$

$\lambda_3 = 7$

$$\begin{bmatrix} -8 & 2 & 0 \\ 0 & -4 & 4 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$-8x_1 + 2x_2 = 0$

$2x_2 = 8x_1$ $\quad \begin{bmatrix} x_1 \\ 4x_1 \\ 4x_1 \end{bmatrix}$

$x_2 = 4x_1$

$x_2 = x_3$

Spektrum yarıçapı

$\max\{|\lambda_1|, |\lambda_2|, |\lambda_3|\} = 7$